

Amitava Shee

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Engineering Leadership | AI, Personalization, Search, Data & Platform Engineering

15+ years leading AI, ML, search, recommendation, big data, cloud, and platform engineering teams across media, fintech, health-care, education, and financial services. Built internet-scale personalization platforms, modernized search, Spark, data lake, and MLOps systems, led engineering organizations, and delivered measurable cost, scale, and product impact.

Selected Impact

- Launched a SiriusXM recommendation engine for 30M consumers on a new streaming platform in 13 months.
- Reduced monthly cloud spend 68%, from \$1.57M to \$507K, without impacting system performance.
- Led a 20-person engineering team operating ML and big data platforms for online and batch inference processing over 1B records daily.
- Built AWS Step Functions, Lambda, and DynamoDB data pipelines for large-scale recommendation workloads.
- Built Spark-oriented data pipelines and data lake/lakehouse capabilities at Ithaka, aligned with Databricks operating patterns.
- Built MLOps pipelines, advanced search ranking, A/B testing, and distributed search platforms on AWS EKS and Apache Solr.

Recent Leadership Experience

SiriusXM | Consultant, Services & Platform Engineering

Nov 2022 - Present | New York, NY

- As acting Director of Personalization, engineered and launched a new recommendation engine for 30M consumers on a new streaming platform, achieving a market launch in a record 13 months.
- Optimized cloud costs by 68%, reducing monthly spend from \$1.57M to \$507K without impacting system performance.
- Led a 20-person engineering team to build and operate a machine learning and big data platform for online and batch inference, processing over 1B records daily at scale.
- Built serverless data pipelines using AWS Step Functions, Lambda, and DynamoDB for large-scale recommendation and personalization workloads.

reThought Insurance | Chief Technology Officer

May 2021 - Mar 2022 | Broomfield, CO

- Led engineering teams across Software Development (9), Data Science (3), and DevOps (3) to build, operationalize, and support a new flood insurance underwriting platform.
- Designed and implemented a cloud-native platform on a self-hosted Kubernetes cluster, integrating with multiple third-party API services and risk analytics platforms.

Ithaka | Principal Engineer | AI & Machine Learning

Jan 2017 - May 2021 | Ann Arbor, MI

- Drove product growth by implementing advanced search features, including a Learning to Boost (L2B) ML model to dynamically generate search engine parameters.
- Developed a continuous MLOps pipeline to data mine clickstream data, train models, and deploy them to production, accelerating the release cycle.
- Built Spark-based data pipelines and data lake/lakehouse capabilities, including Databricks-aligned platform practices for large-scale analytics and ML workflows.

Earlier Experience Summary

Held architecture and engineering leadership roles at University of Michigan (Health System and Medical School), Cognizant Technology Solutions, CONTEXTWEB, Techna International, CA Technologies, Tata Elxsi, Fujitsu ICIM. Work included healthcare research platforms, enterprise architecture, cloud adoption, Epic implementation, banking integrations, securities platforms, SSO, mainframe modernization, EDI, and C/C++ systems engineering.

Core Skills

Engineering Leadership | Artificial Intelligence (AI) | Recommendation Systems | Personalization | Large Language Models (LLM) | Retrieval-Augmented Generation (RAG) | Machine Learning | Machine Learning Operations (MLOps) | Apache Spark | Databricks | Data Lakes | Search Engines | Cloud Architecture | Amazon Web Services (AWS)

Education & Certifications

Birla Institute of Technology | B.Eng. Computer Science

MOR Leaders Program | Scrum Alliance